A. Changelog

Here we list changes between this version (https://arxiv.org/abs/2112.10752v2) of the paper and the previous version, *i.e.* https://arxiv.org/abs/2112.10752v1.

- We updated the results on text-to-image synthesis in Sec. 4.3 which were obtained by training a new, larger model (1.45B parameters). This also includes a new comparison to very recent competing methods on this task that were published on arXiv at the same time as ([59, 109]) or after ([26]) the publication of our work.
- We updated results on class-conditional synthesis on ImageNet in Sec. 4.1, Tab. 3 (see also Sec. D.4) obtained by retraining the model with a larger batch size. The corresponding qualitative results in Fig. 26 and Fig. 27 were also updated. Both the updated text-to-image and the class-conditional model now use classifier-free guidance [32] as a measure to increase visual fidelity.
- We conducted a user study (following the scheme suggested by Saharia et al [72]) which provides additional evaluation for our inpainting (Sec. 4.5) and superresolution models (Sec. 4.4).
- Added Fig. 5 to the main paper, moved Fig. 18 to the appendix, added Fig. 13 to the appendix.

B. Detailed Information on Denoising Diffusion Models

Diffusion models can be specified in terms of a signal-to-noise ratio $\mathrm{SNR}(t) = \frac{\alpha_t^2}{\sigma_t^2}$ consisting of sequences $(\alpha_t)_{t=1}^T$ and $(\sigma_t)_{t=1}^T$ which, starting from a data sample x_0, define a forward diffusion process q as

$$q(x_t|x_0) = \mathcal{N}(x_t|\alpha_t x_0, \sigma_t^2 \mathbb{I}) \tag{4}$$

with the Markov structure for $s < t$:

$$q(x_t|x_s) = \mathcal{N}(x_t|\alpha_{t|s} x_s, \sigma_{t|s}^2 \mathbb{I}) \tag{5}$$

$$\alpha_{t|s} = \frac{\alpha_t}{\alpha_s} \tag{6}$$

$$\sigma_{t|s}^2 = \sigma_t^2 - \alpha_{t|s}^2 \sigma_s^2 \tag{7}$$

Denoising diffusion models are generative models $p(x_0)$ which revert this process with a similar Markov structure running backward in time, *i.e.* they are specified as

$$p(x_0) = \int_z p(x_T) \prod_{t=1}^{T} p(x_{t-1}|x_t) \tag{8}$$

The evidence lower bound (ELBO) associated with this model then decomposes over the discrete time steps as

$$-\log p(x_0) \leq \mathbb{KL}(q(x_T|x_0)|p(x_T)) + \sum_{t=1}^{T} \mathbb{E}_{q(x_t|x_0)} \mathbb{KL}(q(x_{t-1}|x_t, x_0)|p(x_{t-1}|x_t)) \tag{9}$$

The prior $p(x_T)$ is typically choosen as a standard normal distribution and the first term of the ELBO then depends only on the final signal-to-noise ratio $\mathrm{SNR}(T)$. To minimize the remaining terms, a common choice to parameterize $p(x_{t-1}|x_t)$ is to specify it in terms of the true posterior $q(x_{t-1}|x_t, x_0)$ but with the unknown x_0 replaced by an estimate $x_\theta(x_t, t)$ based on the current step x_t. This gives [45]

$$p(x_{t-1}|x_t) \coloneqq q(x_{t-1}|x_t, x_\theta(x_t, t)) \tag{10}$$

$$= \mathcal{N}(x_{t-1}|\mu_\theta(x_t, t), \sigma_{t|t-1}^2 \frac{\sigma_{t-1}^2}{\sigma_t^2} \mathbb{I}), \tag{11}$$

where the mean can be expressed as

$$\mu_\theta(x_t, t) = \frac{\alpha_{t|t-1} \sigma_{t-1}^2}{\sigma_t^2} x_t + \frac{\alpha_{t-1} \sigma_{t|t-1}^2}{\sigma_t^2} x_\theta(x_t, t). \tag{12}$$

In this case, the sum of the ELBO simplify to

$$\sum_{t=1}^{T} \mathbb{E}_{q(x_t|x_0)} \mathbb{KL}(q(x_{t-1}|x_t, x_0)|p(x_{t-1}) = \sum_{t=1}^{T} \mathbb{E}_{\mathcal{N}(\epsilon|0,\mathbb{I})} \frac{1}{2}(\text{SNR}(t-1) - \text{SNR}(t))\|x_0 - x_\theta(\alpha_t x_0 + \sigma_t \epsilon, t)\|^2 \tag{13}$$

Following [30], we use the reparameterization

$$\epsilon_\theta(x_t, t) = (x_t - \alpha_t x_\theta(x_t, t))/\sigma_t \tag{14}$$

to express the reconstruction term as a denoising objective,

$$\|x_0 - x_\theta(\alpha_t x_0 + \sigma_t \epsilon, t)\|^2 = \frac{\sigma_t^2}{\alpha_t^2} \|\epsilon - \epsilon_\theta(\alpha_t x_0 + \sigma_t \epsilon, t)\|^2 \tag{15}$$

and the reweighting, which assigns each of the terms the same weight and results in Eq. (1).

C. Image Guiding Mechanisms

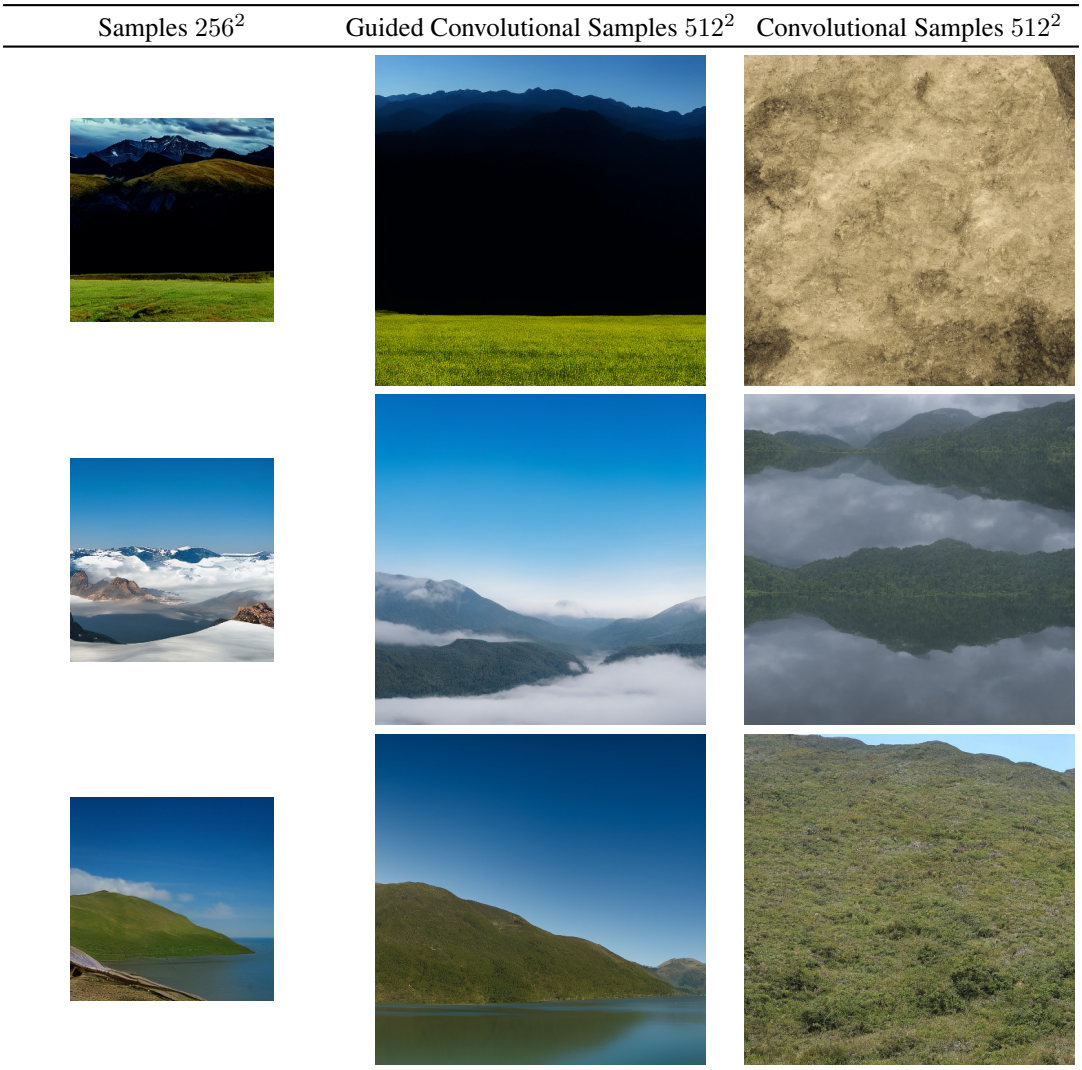

Figure 14. On landscapes, convolutional sampling with unconditional models can lead to homogeneous and incoherent global structures (see column 2). L_2-guiding with a low resolution image can help to reestablish coherent global structures.

An intriguing feature of diffusion models is that unconditional models can be conditioned at test-time [15, 82, 85]. In particular, [15] presented an algorithm to guide both unconditional and conditional models trained on the ImageNet dataset with a classifier $\log p_\Phi(y|x_t)$, trained on each x_t of the diffusion process. We directly build on this formulation and introduce post-hoc *image-guiding*:

For an epsilon-parameterized model with fixed variance, the guiding algorithm as introduced in [15] reads:

$$\hat{\epsilon} \leftarrow \epsilon_\theta(z_t, t) + \sqrt{1-\alpha_t^2}\, \nabla_{z_t} \log p_\Phi(y|z_t)\ . \tag{16}$$

This can be interpreted as an update correcting the "score" ϵ_θ with a conditional distribution $\log p_\Phi(y|z_t)$.

So far, this scenario has only been applied to single-class classification models. We re-interpret the guiding distribution $p_\Phi(y|T(\mathcal{D}(z_0(z_t))))$ as a general purpose image-to-image translation task given a target image y, where T can be any differentiable transformation adopted to the image-to-image translation task at hand, such as the identity, a downsampling operation or similar.